Question and design

Does model width improve contact accuracy at the same one-epoch exposure? Baseline: exp277, native plus MPNN corpus, 24 layers. Hidden width 2048 -> 4096; MLP width 8192 -> 16384. Keep 32 query / 8 KV heads; head width 64 -> 128. Keep data, optimizer schedule, seeds and global batch fixed.

Architecture and smoke verified

5,861,984,256 parameters, versus exp277's 1,471,374,336 (3.984x). Architecture, tokenizer, data and recovery checks passed. 64-H100 smoke: ten updates, train loss 8.56948, eval loss 8.03575. 2.821 seconds/update; native and HF checkpoints saved with tokenizer. Compressed smoke schedule was spiky; this is an execution check.

Current status — October 26

Both a04 jobs remain running on four nodes / 32 H100s each, at batch priority. Original: step 117,186, 44.00% complete, LR 0.001; numerically diverged. Fork: step 116,027, 43.56% complete, LR 0.000707; stable. Original first training loss above 4 in the inspected window: step 77,229. Original validation rose from 3.09974 at 76,104 to 11.88947 at 78,218. It peaked at 42.43565 and remains 17.96501 at step 116,270. Both remain active as requested. Recommend stopping the divergent control. Fork final cooldown still starts at 213,076; total horizon 266,345 updates.

TPU trial succeeded

Interactive v5p-64: 32 chips across 8 VMs; all 20 updates completed. Finished October 16; no worker failures or preemptions. Native and HF step-19 checkpoints verified; tokenizer included. Median steps 5–19: 7.40 s/update, about 32% longer than GPU steps. Short native-only smoke; throughput is preliminary, not cost-normalized. Production continues on GPUs. No bulk transfer or migration performed.

Matched validation: lower LR gives a small advantage

The fork beats the default at all 19 exact-step post-fork evaluations. Same native + MPNN corpus, global batch, validation and absolute data position. At step 114,156: default 3.071639; fork 3.066603; original 17.649031. Fork minus default: -0.005036 nats/token, or 0.50% lower perplexity. This reverses the original wide model's disadvantage before the fork. Reducing LR is supported by stability and loss for this particular run. Single seed, incomplete epoch; width and LR differ from the smaller default. No contact-accuracy evaluation for the wider model yet.

Other architectures: context, not a common leaderboard

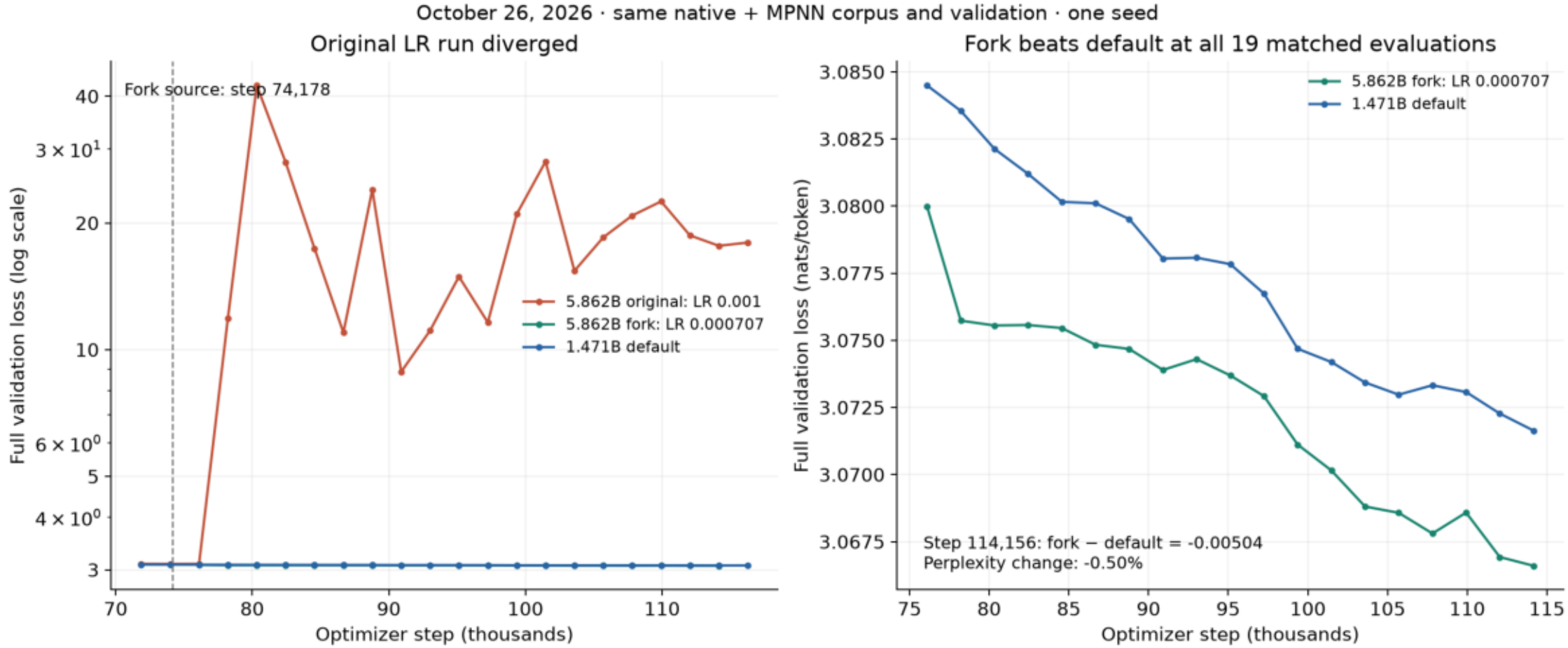
exp108 depth-scaled 2.93B: all three LR trials unstable around 6.8k updates. exp108 width-scaled 2.78B: all three trials finished; older recipe and validation. exp262 NoPE + smear: final loss 3.00207 versus its control's 2.97454. Its early advantage reversed, illustrating why interim rankings can mislead. exp157 learned RoPE deltas: corrected CE 3.0654 versus vanilla 3.0815. Only exp277 provides the same native + MPNN recipe as the current wider run. Raw historical validation losses cannot be pooled across cache/logging paths.

Parallel LR fork — October 22

Original schedule retains its 0.001 plateau and final cooldown. Fork restores checkpoint 74,178, resuming at absolute step 74,179. Weights, Adam moments/counters, RNG, data position and augmentation are preserved. Linear reduction to 0.000707 over 2,114 updates; target reached at step 76,293. Hold until original final cooldown at step 213,076; final LR target 0.0000707. Same 266,345-update horizon, separate W&B and checkpoint identities. Source copy and counters verified; 14 local configuration/restore/serialization tests passed. October 22 a02 failed before training; October 23 credentials refreshed. Retry a03 exposed CLI-class serialization; moved optimizer into an importable module. Corrected a04 is training; first nine LRs match the schedule within 1.14e-10. First loss at 74,179 exactly matches the original, confirming the starting comparison.

lr_fork_comparison.png

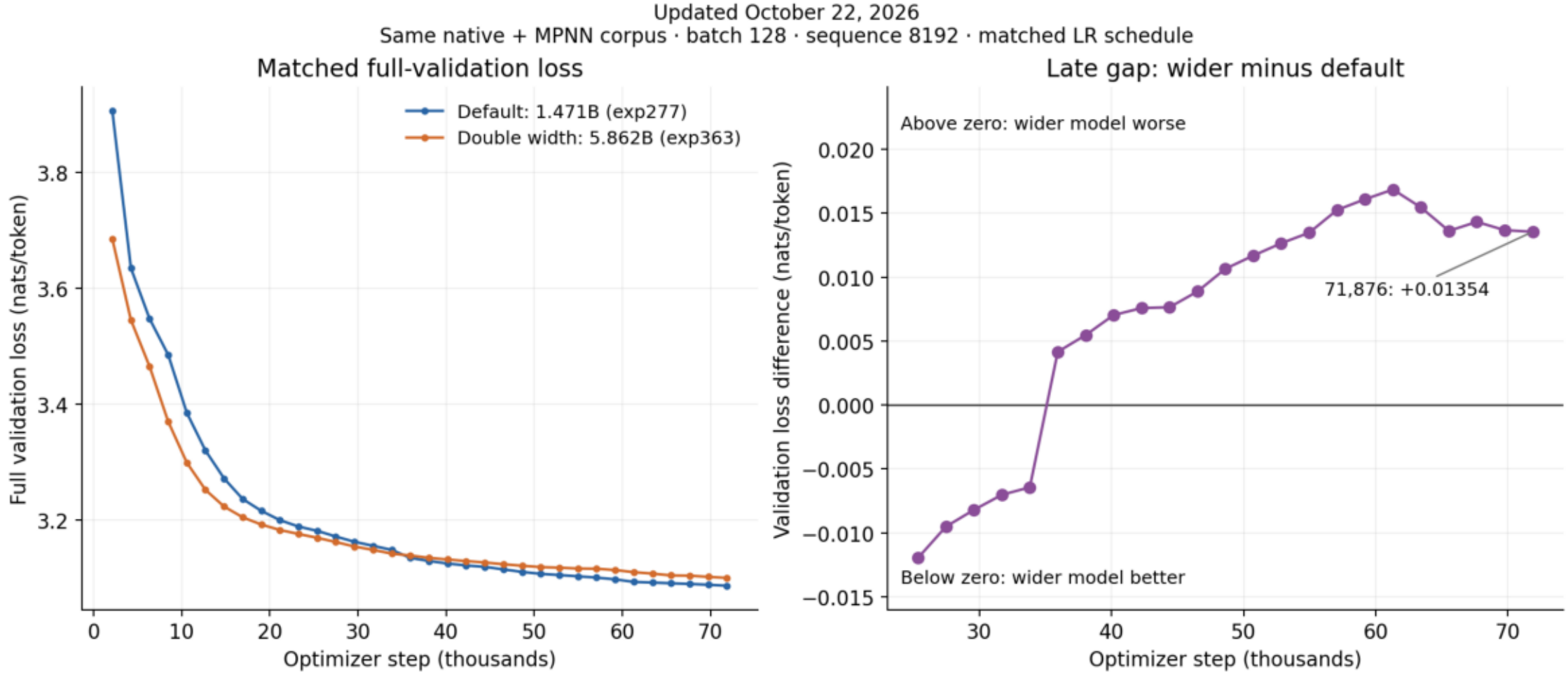
W&B snapshot 2026-10-26T15:55:10.854315+00:00. Original high-LR divergence is shown on a log scale at left; right compares fork/default at exact matching steps. No smoothing or interpolation. LR reaches 0.000707 at step 76,293. The small validation advantage does not establish contact accuracy.



$ plot_lr_fork.py

training_comparison.png

Full-validation observations through step 71,876, exported October 22, 2026. Exact-step comparison; no interpolation or smoothing. Same logical training recipe, different hardware/gradient accumulation; single seed, incomplete epoch. The right panel zooms into observations after step 25,000.




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$ plot_training_comparison.py
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